Supplemental Material for "Sign memory amplifies counting noise in market order flow"

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HIGHER CUMULANTS DO NOT DEFINE A GAUSSIAN CROSSOVER

For each asset and dyadic counting window $T = 1, 2, \ldots, 128$ hours, we form non-overlapping accumulated flows Q_T and report the standardised third and fourth cumulants

$$\gamma_1(T) = \frac{\kappa_3(T)}{\kappa_2(T)^{3/2}}, \qquad \gamma_2(T) = \frac{\kappa_4(T)}{\kappa_2(T)^2}. \tag{1}$$

The 95% intervals use 999 circular moving-block bootstrap replicates. The block length corresponds to four weeks of clock time at every T, so the resampling retains short- and intermediate-range dependence within each block.

Neither statistic approaches zero uniformly across assets [Fig. 1 and Table I]. BTC loses most of its excess kurtosis by $T = 128$ but remains skewed; ETH becomes markedly more skewed and heavy-tailed; BNB and SOL retain nonzero excess kurtosis. We therefore do not define a characteristic Gaussian scale T^*. This does not alter the algebraic statement in the Letter: Gaussianity is a sufficient null explanation for a linear symmetry function, not an empirical description we impose on all windows. Conversely, a linear $\log[P(Q)/P(-Q)]$ can occur for a non-Gaussian exponentially tilted distribution, so linearity alone does not establish either Gaussianity or a nontrivial fluctuation theorem.

PARALLEL FLOW DEFINITIONS AND SIGN–MAGNITUDE ATTRIBUTION

The primary analysis is repeated without mixing observables. Raw signed base volume is $\varepsilon_t^{(r)} = 2V_t^+ - V_t$; normalised imbalance is $\varepsilon_t^{(n)} = \varepsilon_t^{(r)}/V_t$. Within each quarter–hour-of-week stratum we remove the corresponding mean. The joint null permutes the complete residual. Writing that residual as $s_t m_t$, the sign null permutes s_t while keeping m_t at its observed times, and the magnitude null permutes m_t while keeping s_t fixed. Each surrogate is re-centred within stratum before weekly aggregation.

Table II gives all results. The sign null is close to the joint null, whereas the magnitude null has substantially larger variance. Equivalently, retaining sign order recovers more of the observation than retaining magnitude order. The comparison is not additive: both attribution nulls break contemporaneous sign–magnitude pairing, so their difference can include coupling between signs and sizes.

BTC, ETH, BNB, and SOL form the primary panel. After fixing the analysis, we added XRP, ADA, DOGE, and AVAX as a post-hoc robustness panel with the same four-year hourly coverage. In the extension, the joint-null ratios span 2.08–5.67 for raw flow and 2.83–5.64 for normalised flow. The sign null remains close to the joint null in every added asset, while the magnitude null is smaller. This replication broadens the empirical scope, but it is not part of the predeclared four-asset decision rule.

Uncertainty in the observed weekly variance is estimated with a circular moving-block bootstrap over the weekly sums. We use 3999 replicates and report 4-, 8-, 13-, and 26-week block sensitivity. These intervals are conditional on the median empirical null; they do not include Monte Carlo uncertainty in that median, which is negligible relative to the weekly-sample uncertainty at 9999 permutations. The eight-week intervals appear in Table II and Fig. 1 of the Letter.

SENSITIVITY TO THE TEMPORAL CONDITIONING OF THE NULL

The primary joint null permutes signed flow within (calendar quarter, hour of week) strata. We repeat the calculation with calendar-month strata and with consecutive 14-day blocks anchored on the first complete Monday in the sample. In every case the mean is removed within the same strata used for permutation. Calendar edge strata containing one retained candle are left fixed; their fraction is reported in Table III.

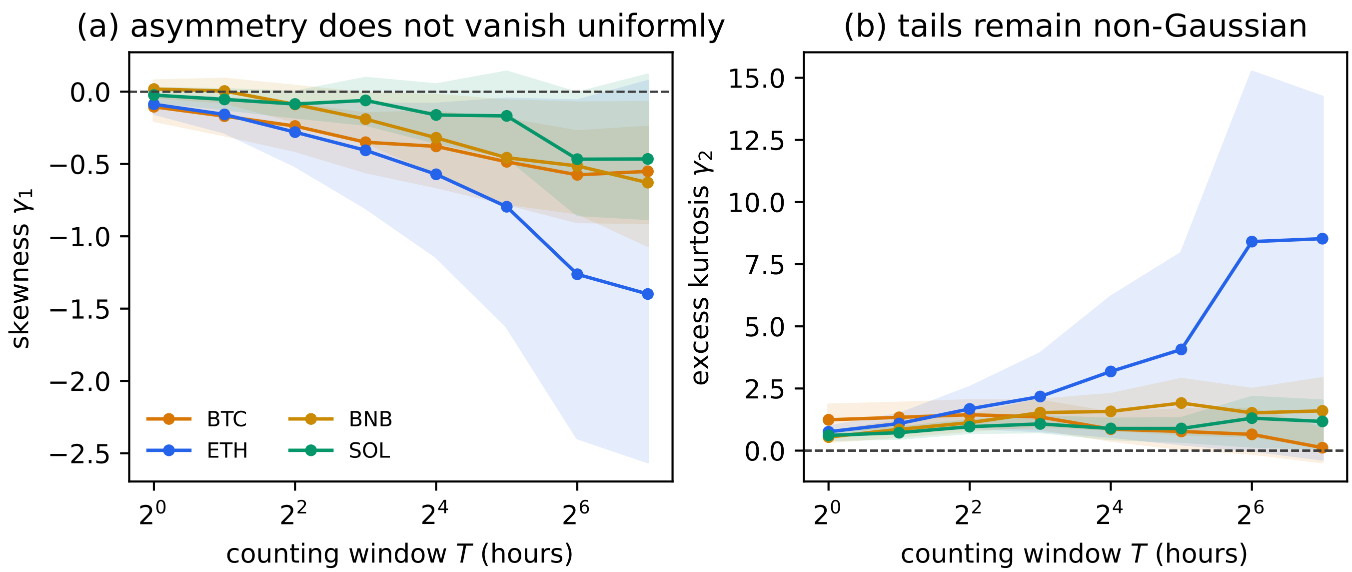

FIG. 1. Scale dependence of (a) skewness and (b) excess kurtosis. Shading is the moving-block bootstrap 95% interval. The zero line is Gaussian.

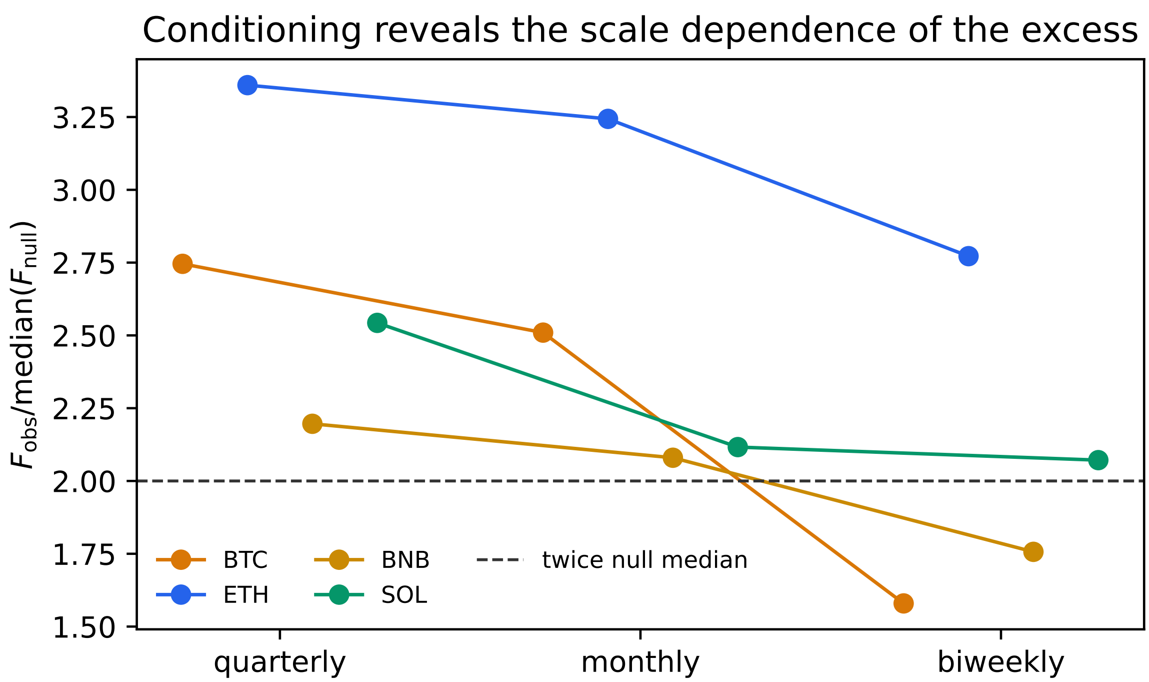

FIG. 2. Observed weekly variance relative to the median of each conditioned null. The dashed line marks a factor of two.

Table III reports both flow definitions. Finer conditioning removes part of the low-frequency organisation whose contribution the quarterly null measures. The sensitivity therefore identifies scale content in the excess and rules out partition invariance.

The original predeclared screen was $V_{\mathrm{obs}} > 2Q_{0.9875}(V_{\mathrm{null}})$ for all four assets under the raw-flow quarterly joint null. BNB failed, so the all-assets gate failed. The later normalised-flow analysis is a robustness analysis and is not substituted into that decision rule. Pairwise weekly-flow correlations are 0.14–0.39 for raw flow and −0.13–0.26 for normalised flow within the primary panel. Over all eight assets the corresponding ranges are 0.01–0.45 and −0.16–0.27; neither panel should be treated as a collection of independent replications.

WAVELET CONTROL OF MEAN DRIFT

The wavelet estimators are applied directly to the hourly signed-flow series, not to a separately integrated profile. A Daubechies wavelet with p vanishing moments annihilates polynomial trends of degree smaller than p [1, 2]. Thus db1 (Haar) removes a constant mean, db2 a linear trend, and db3 a quadratic trend.

The implementation is calibrated on fractional Gaussian noise with known $\nu = 1.40$. The block-variance, db1, db2 and db3 estimates agree within 0.01 without drift. Injecting linear or quadratic trends inflates the block estimate while leaving the appropriate wavelet estimates fixed. On market data all three wavelet estimates remain superdiffusive, $\nu = 1.10$–1.36 [Table IV], but their local slopes do not resolve the monotone crossover seen in the drift-sensitive block variance. This establishes global superdiffusion while leaving the local crossover undecided.

SCALING MODEL REASSESSMENT

For $C(\tau) \sim \tau^{-a}$, the map $\nu = 2 - a$ applies only for $0 < a < 1$. At $a = 1$ the leading form is $T \log T$, and for $a > 1$ it is T. Applying this map asset by asset gives Table V. The single broad-window power-law model misses all eight block-variance exponents by more than three reported block-fit standard errors, although those discrepancies do not include uncertainty in the ACF prediction. The two-regime intervals are weaker still: they omit amplitudes and crossover uncertainty, and BNB lies outside its band. We therefore use neither construction as independent confirmation. The exact finite-sample covariance identity already fixes V_T from $C(\tau)$; the robust empirical statement is global superdiffusion under drift-resistant estimators.

TABLE I. Standardised cumulants at the endpoints of the dyadic scale grid. Intervals in Fig. 1 use the full grid.

	$\gamma_1(1)$	$\gamma_1(128)$	$\gamma_2(1)$	$\gamma_2(128)$
BTC	-0.10	-0.55	1.24	0.12
ETH	-0.09	-1.40	0.76	8.53
BNB	0.02	-0.63	0.54	1.60
SOL	-0.02	-0.47	0.61	1.17

TABLE II. Decomposition of weekly variance amplification under quarterly $\times$ hour-of-week conditioning. R_J, R_S, and R_M divide observed variance by the median joint-order, sign-order, and magnitude-order null. The interval is an eight-week moving-block bootstrap interval for R_J, conditional on its null median.

asset	flow	R_J	R_S	R_M	95% interval for R_J
BTC	raw	2.75	2.82	2.14	[1.65,4.26]
BTC	normalized	3.20	3.35	1.64	[2.32,4.14]
ETH	raw	3.36	3.43	2.10	[2.43,4.38]
ETH	normalized	4.22	4.43	1.89	[2.90,6.11]
BNB	raw	2.19	2.29	1.66	[1.71,2.72]
BNB	normalized	4.20	4.39	1.86	[3.05,5.53]
SOL	raw	2.55	2.61	1.75	[1.95,3.19]
SOL	normalized	2.99	3.14	1.52	[2.22,3.81]
XRP	raw	5.67	5.81	1.81	[3.10,9.10]
XRP	normalized	5.64	5.93	1.87	[4.13,7.33]
ADA	raw	2.08	2.10	1.75	[1.50,2.72]
ADA	normalized	2.83	2.98	1.59	[1.99,3.83]
DOGE	raw	3.26	3.29	2.11	[1.84,5.37]
DOGE	normalized	3.99	4.19	1.71	[3.06,4.98]
AVAX	raw	2.36	2.53	1.57	[1.56,3.30]
AVAX	normalized	3.18	3.35	1.55	[2.40,4.02]

TABLE III. Sensitivity of the joint weekly variance null. $R_{50} = V_{\mathrm{obs}}/\mathrm{median}(V_{\mathrm{null}})$ and $R_{98.75} = V_{\mathrm{obs}}/Q_{0.9875}(V_{\mathrm{null}})$. Primary quarterly rows use 9999 permutations; monthly and biweekly rows use 1999.

asset	flow	conditioning	R_{50}	$R_{98.75}$	p_{emp}	fixed fraction
BTC	raw	quarterly	2.75	2.08	0.0001	0.00%
BTC	raw	monthly	2.52	1.83	0.0005	0.14%
BTC	raw	biweekly	1.57	1.07	0.0070	0.97%
BTC	normalized	quarterly	3.20	2.48	0.0001	0.00%
BTC	normalized	monthly	2.67	2.05	0.0005	0.14%
BTC	normalized	biweekly	2.27	1.59	0.0005	0.97%
ETH	raw	quarterly	3.36	2.63	0.0001	0.00%
ETH	raw	monthly	3.23	2.41	0.0005	0.14%
ETH	raw	biweekly	2.79	1.87	0.0005	0.97%
ETH	normalized	quarterly	4.22	3.35	0.0001	0.00%
ETH	normalized	monthly	3.54	2.78	0.0005	0.14%
ETH	normalized	biweekly	2.84	2.09	0.0005	0.97%
BNB	raw	quarterly	2.19	1.69	0.0001	0.00%
BNB	raw	monthly	2.07	1.54	0.0005	0.14%
BNB	raw	biweekly	1.75	1.17	0.0015	0.97%
BNB	normalized	quarterly	4.20	3.29	0.0001	0.00%
BNB	normalized	monthly	3.59	2.81	0.0005	0.14%
BNB	normalized	biweekly	3.35	2.44	0.0005	0.97%
SOL	raw	quarterly	2.55	2.02	0.0001	0.00%
SOL	raw	monthly	2.12	1.63	0.0005	0.14%
SOL	raw	biweekly	2.10	1.45	0.0005	0.97%
SOL	normalized	quarterly	2.99	2.36	0.0001	0.00%
SOL	normalized	monthly	2.20	1.73	0.0005	0.14%
SOL	normalized	biweekly	2.31	1.68	0.0005	0.97%

TABLE IV. Global exponent ν over $T = 8, \ldots, 256$. db1 removes a constant, db2 a linear trend, and db3 a quadratic trend.

asset	block variance	db1	db2	db3
BTC	1.42	1.20	1.14	1.13
ETH	1.37	1.24	1.26	1.29
BNB	1.43	1.35	1.36	1.30
SOL	1.31	1.15	1.13	1.10

TABLE V. Asset-specific scaling reassessment. The consistency band maps $C(\tau) \sim \tau^{-a}$ to $\nu = 2 - a$ only for $a < 1$; $a > 1$ maps to diffusion. Amplitudes are not fitted, so the band is not a quantitative prediction.

asset	a_f	a_s	consistency band	ν_{1h}	ν_{4h}
BTC	1.00	0.17	[1.00,1.83]	1.33	1.32
ETH	0.87	0.22	[1.13,1.78]	1.32	1.31
BNB	0.46	0.29	[1.54,1.71]	1.37	1.37
SOL	1.71	0.16	[1.00,1.84]	1.25	1.22

[1] I. Daubechies, *Ten Lectures on Wavelets* (SIAM, Philadelphia, 1992).
[2] D. B. Percival and A. T. Walden, *Wavelet Methods for Time Series Analysis* (Cambridge University Press, Cambridge, 2000).